A Feng–Tian–Wei-type central limit theorem for the high-temperature SYK partition function

Abstract

Fix an even interaction order $q \geq 4$ and let $Z_N(\beta)$ be the normalized-trace partition function of the Gaussian SYK model. We prove, for all sufficiently small fixed real $\beta \neq 0$,

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1).$$

The proof is a high-temperature modification of the argument of Feng–Tian–Wei. Their fixed-test-function proof isolates the radial statistic $M_N^{-1}\sum_I (g_I^2 - 1)$ by subtracting a multiple of x^2. For the N-dependent test function representing the partition function, we replace that asymptotic subtraction by an exact second-Wiener-chaos projection. The remaining chaoses are controlled in two ranges: an FTW-type parity count up to order $A \log N$, and analytic suppression above $A \log N$. The only external model-specific inputs are a published complex second-moment estimate and the published high-temperature annealed free-energy limit; every deduction from those two inputs is included below.

1 Statement, normalization, and external inputs

1.1 The SYK model

Fix an even integer $q \geq 4$, and let N be even. Let

$$\mathcal{I}_N = \{I \subset [N] : |I| = q\}, \qquad M_N = \binom{N}{q}, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}},$$

and let $(g_I)_{I \in \mathcal{I}_N}$ be independent standard Gaussian random variables. Let $\psi_1, \ldots, \psi_N$ be Majorana matrices satisfying $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$. For $I = \{i_1 < \cdots < i_q\}$ set

$$\psi_I = i^{q/2}\psi_{i_1} \cdots \psi_{i_q}.$$

Thus ψ_I is Hermitian, $\psi_I^2 = \mathbb{I}$, and $\|\psi_I\| = 1$. Put

$$H_N(g) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad Z_N(z; g) = \operatorname{tr} e^{-zH_N(g)}, \qquad F_N(z) = \mathbb{E}Z_N(z; g).$$

Here $\operatorname{tr} = 2^{-N/2}\operatorname{Tr}$ is the normalized trace, so $Z_N(0) = F_N(0) = 1$.

It is useful to record the normalization used by Feng–Tian–Wei. Define

$$\widehat{H}_N(g) = M_N^{-1/2} \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad \tau_N = \sigma_N \sqrt{M_N}. \tag{1}$$

Then $H_N = \tau_N \widehat{H}_N$ and

$$\tau_N^2 = \sigma_N^2 M_N = \frac{N}{q}\big(1 + O_q(N^{-1})\big). \tag{2}$$

Consequently, if $\widehat{L}_N(f) = \operatorname{tr} f(\widehat{H}_N)$, then

$$Z_N(\beta) = \widehat{L}_N(f_{N,\beta}), \qquad f_{N,\beta}(x) = e^{-\beta\tau_N x}. \tag{3}$$

1.2 External input I: a complex second-moment estimate

We use the following published high-temperature input. It is Theorems 3 and 4 of [2], with the deterministic change of inverse-temperature coming from the Majorana normalization. In particular, the constants below depend only on q.

Proposition 1.1 (Complex second-moment input)**.** *There are constants $R_q > 0$, $C_q < \infty$, and $N_q < \infty$ such that, for every even $N \geq N_q$,*

$$F_N(z) \neq 0 \qquad (|z| \leq R_q), \tag{4}$$

and

$$\sup_{|z|\leq R_q} \mathbb{E}\left|\frac{Z_N(z;g)}{F_N(z)} - 1\right|^2 \leq C_q N^{1-q/2}. \tag{5}$$

Proof. This is Theorems 3 and 4 of [2], specialized to the unperturbed model, followed only by a deterministic normalization change. That paper uses Majoranas satisfying $\{\psi_i^{\mathrm{Zlo}}, \psi_j^{\mathrm{Zlo}}\} = \delta_{ij}\mathbb{I}$. Thus $\psi_i^{\mathrm{Zlo}} = 2^{-1/2}\psi_i$, $H_N^{\mathrm{Zlo}} = 2^{-q/2}H_N$, and

$$Z_N^{\mathrm{ours}}(z) = Z_N^{\mathrm{Zlo}}(2^{q/2}z).$$

Normalized versus unnormalized trace cancels in the ratio in (5). We choose R_q so that $\{|2^{q/2}z| \leq R_q^{\mathrm{Zlo}}\}$ lies inside the zero-free and second-moment region in those theorems. This gives (4)–(5) for all sufficiently large N. □

Set

$$a = \frac{q}{2} - 1 > 0, \qquad Y_N(z;g) = \frac{Z_N(z;g)}{F_N(z)}.$$

Thus $\mathbb{E}Y_N(z) = 1$, and the right-hand side of (5) is $C_q N^{-a}$. Let P_r denote the orthogonal projection in complex $L^2(\gamma_N)$ onto the r-th homogeneous Gaussian chaos, where γ_N denotes standard Gaussian measure on $\mathbb{R}^{M_N}$. We write $\|X\|_2^2 = \mathbb{E}|X|^2$.

Lemma 1.2 (L^2-valued analyticity)**.** *For each fixed N, the map $z \mapsto Z_N(z;\cdot)$ is entire with values in $L^2(\gamma_N)$. Hence $z \mapsto Y_N(z;\cdot)$ is $L^2(\gamma_N)$-valued analytic on every zero-free domain of F_N.*

Proof. Since $\|\psi_I\| = 1$,

$$\|H_N(g)\| \leq \sigma_N \sum_{I\in\mathcal{I}_N} |g_I|.$$

For $|z| \leq R$ and every $k \geq 0$, the spectral theorem gives

$$|\partial_z^k Z_N(z;g)| \leq \|H_N(g)\|^k \, e^{R\|H_N(g)\|}.$$

The square of the right-hand side is integrable because the Gaussian vector is finite dimensional. Dominated convergence therefore permits differentiation in L^2 on every compact disk. Division by the nonvanishing deterministic analytic function F_N proves the second assertion. □

2 The precise modification of the FTW argument

For a fixed test function f, Feng–Tian–Wei [1] prove the Gaussian SYK linear-statistics CLT by isolating

$$Q_N := \operatorname{tr} \widehat{H}_N^2 - \mathbb{E}\operatorname{tr} \widehat{H}_N^2 = \frac{1}{M_N}\sum_{I\in\mathcal{I}_N}(g_I^2 - 1). \tag{6}$$

In the fixed-q regime, their coefficient is

$$A(f) = \int_{\mathbb{R}} \frac{x f'(x)}{2} \, \rho_\infty(\mathrm{d}x),$$

and their proof amounts to showing that the centered statistic for $f - A(f)x^2$ has variance $o(M_N^{-1})$. Since $\sqrt{M_N} Q_N \Rightarrow \mathcal{N}(0,2)$, the radial term gives their limit.

There are two obstructions to substituting (3) directly into that theorem. First, $f_{N,\beta}$ depends on N through $\tau_N \asymp \sqrt{N}$. Second, neither the fixed-function polynomial approximation nor the bounded-derivative estimate in [1] is uniform for this sequence. In particular, the theorem as stated in [1] does *not* imply a CLT for $Z_N(\beta)$.

The modification is to work with the relative partition function Y_N and to replace the limiting coefficient $A(f)$ by the exact finite-N coefficient

$$A_N(\beta) := \frac{\beta}{2} \, \partial_\beta \log F_N(\beta). \tag{7}$$

Indeed, the next lemma proves the exact FTW decomposition

$$Y_N(\beta) - 1 = A_N(\beta) Q_N + \mathcal{R}_N(\beta), \qquad \mathcal{R}_N = \sum_{r \geq 3} P_r Y_N. \tag{8}$$

The new estimate, replacing the fixed-test-function variance computation of FTW, is

$$\mathbb{E}|\mathcal{R}_N(\beta)|^2 = o(N^2/M_N) = o(N^{2-q}). \tag{9}$$

Sections 4–6 prove precisely (9).

3 Exact finite-N FTW projection

Lemma 3.1 (First and second chaoses)**.** *For every real β,*

$$P_1 Y_N(\beta) = 0$$

and

$$P_2 Y_N(\beta) = \frac{\beta \, \partial_\beta \log F_N(\beta)}{2M_N} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{10}$$

Equivalently, since $\operatorname{tr} H_N^2 = \sigma_N^2 \sum_I g_I^2$, the second chaos is the L^2 projection of $Y_N - 1$ onto the centered radial statistic $\operatorname{tr} H_N^2 - \mathbb{E} \operatorname{tr} H_N^2$.

Proof. For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$, define $\chi_s(I) = \prod_{x \in I} s_x$. The Clifford sign-flip automorphism is unitarily implemented, and hence

$$Z_N(\beta; g) = Z_N(\beta; (\chi_s(I) g_I)_I).$$

The corresponding parity rule for Hermite coefficients kills the first chaos and every off-diagonal second-chaos monomial $g_I g_J$, $I \neq J$. Permutation symmetry shows that the remaining diagonal coefficients agree. Consequently

$$P_2 Z_N(\beta) = b_N(\beta) \sum_I (g_I^2 - 1)$$

for a deterministic scalar b_N.

The Hermite coefficient formula and Gaussian integration by parts give

$$\begin{aligned} 2M_N b_N(\beta) &= \sum_I \mathbb{E} \, \partial_{g_I}^2 Z_N(\beta) = \sum_I \mathbb{E}\big[g_I \partial_{g_I} Z_N(\beta) \big] \\ &= \beta \, \partial_\beta \mathbb{E} Z_N(\beta) = \beta F_N'(\beta). \end{aligned}$$

Here the third equality is Euler's identity, since H_N is linear in g. Dividing by the deterministic quantity $F_N(\beta)$ proves (10). □

4 Analytic control of the high-order remainder

The next lemma is the step which upgrades the polynomially weak estimate (5) to the precision required for the CLT.

Lemma 4.1 (Analytic suppression of high chaoses)**.** *Let $w \in \mathbb{C}$ satisfy $|w| < R$, and let $G(z)$ be analytic on a neighborhood of $|z| \le R$ with values in a Hilbert orthogonal sum $\mathcal{H} = \bigoplus_{r\ge 1} \mathcal{H}_r$. Write $G_r(z)$ for its projection onto $\mathcal{H}_r$. Suppose that G_r has a zero of order at least r at the origin. Then, for every integer $m \ge 0$,*

$$\sum_{r>m} \|G_r(w)\|_{\mathcal{H}}^2 \le \frac{(|w|/R)^{2(m+1)}}{1-(|w|/R)^2} \frac{1}{2\pi} \int_0^{2\pi} \left\|G(Re^{i\theta})\right\|_{\mathcal{H}}^2 d\theta. \tag{11}$$

Proof. Write

$$G_r(z) = \sum_{k\ge r} z^k v_{r,k}, \qquad v_{r,k} \in \mathcal{H}_r.$$

Parseval's identity in the angular variable gives

$$A_r(R) := \frac{1}{2\pi} \int_0^{2\pi} \left\|G_r(Re^{i\theta})\right\|_{\mathcal{H}}^2 d\theta = \sum_{k\ge r} R^{2k} \|v_{r,k}\|_{\mathcal{H}}^2 .$$

With $t = |w|/R < 1$, Cauchy–Schwarz yields

$$\|G_r(w)\|_{\mathcal{H}}^2 \le \left(\sum_{k\ge r} t^{2k}\right) A_r(R) = \frac{t^{2r}}{1-t^2} A_r(R).$$

Sum over $r > m$, use $t^{2r} \le t^{2(m+1)}$, and then use orthogonality of the spaces $\mathcal{H}_r$ under the angular integral. □

Lemma 4.2 (Vanishing order)**.** *For every $r \ge 1$, the $L^2(\gamma_N)$-valued analytic function $z \mapsto P_r Y_N(z)$ has a zero of order at least r at $z = 0$.*

Proof. The Taylor coefficient of order k of $Z_N(z; g)$ is

$$\frac{(-1)^k}{k!} \operatorname{tr} H_N(g)^k,$$

which is a polynomial of total degree k in the Gaussian vector g. It has no Gaussian-chaos component of order larger than k. Therefore $P_r Z_N(z)$ is divisible by z^r. Since F_N is deterministic, $F_N(0) = 1$, and F_N has no zero on the disk under consideration, division by $F_N(z)$ preserves this vanishing order. □

Apply Lemma 4.1 to $G(z) = Y_N(z) - 1$ and $G_r(z) = P_r Y_N(z)$. Combining Proposition 1.1 with Lemmas 4.1 and 4.2, for every $0 < |\beta| < R < R_q$ and every $m \ge 0$ we obtain

$$\sum_{r>m} \|P_r Y_N(\beta)\|_2^2 \le \frac{C_q}{1-(|\beta|/R)^2} N^{-a} \left(\frac{|\beta|}{R}\right)^{2(m+1)}. \tag{12}$$

5 Uniform control of chaoses up to logarithmic order

Lemma 5.1 (Real-temperature derivative bound)**.** *For real β, every multi-index α of total order r satisfies*

$$\left|\partial_g^\alpha Z_N(\beta; g)\right| \le (|\beta|\sigma_N)^r Z_N(\beta; g). \tag{13}$$

Proof. Label the r differentiations by $I_1, \dots, I_r$. The iterated Duhamel formula is a sum over $r!$ permutations of integrals over the simplex $\Delta_r = \{t_j \geq 0 : \sum_{j=0}^{r} t_j = 1\}$, whose volume is $1/r!$. Every integrand, apart from the factor $(|\beta|\sigma_N)^r$, has the form

$$\operatorname{tr}\left(e^{-t_0\beta H_N}\psi_{I_{\pi(1)}}e^{-t_1\beta H_N}\cdots\psi_{I_{\pi(r)}}e^{-t_r\beta H_N}\right).$$

Noncommutative Hölder, with Schatten exponents $1/t_j$, and $\|\psi_I\| = 1$, bounds its absolute value by

$$\prod_{j=0}^{r}\left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j} = Z_N(\beta; g).$$

The permutation count cancels the simplex volume, proving (13). The same labeled formula covers repeated coordinates. □

Call a multi-index α *even* if

$$\sum_{I\ni x}\alpha_I \equiv 0 \pmod 2 \qquad \text{for every } x \in [N].$$

The sign-gauge symmetry used in Lemma 3.1 shows that only even multi-indices have nonzero Hermite coefficients.

Lemma 5.2 (Logarithmic-order finite-chaos estimate)**.** *For every fixed real β there is $D = D(q,\beta) < \infty$ such that, for all $r \geq 1$ and all N,*

$$\|P_r Y_N(\beta)\|_2^2 \leq \exp\big(Dr\log(r+1)\big)N^{-ar}. \tag{14}$$

Consequently, for every fixed $A < \infty$,

$$\sum_{r=3}^{\lceil A\log N\rceil}\|P_r Y_N(\beta)\|_2^2 = o(N^{-2a}). \tag{15}$$

Proof. The Hermite coefficient formula, Lemma 5.1, and $F_N(\beta) > 0$ give

$$\begin{aligned}\|P_r Y_N(\beta)\|_2^2 &= \frac{1}{F_N(\beta)^2}\sum_{|\alpha|=r}\frac{1}{\alpha!}\left|\mathbb{E}\partial_g^\alpha Z_N(\beta)\right|^2\\ &\leq (|\beta|\sigma_N)^{2r}\sum_{\substack{|\alpha|=r\\ \alpha\text{ even}}}\frac{1}{\alpha!}.\end{aligned} \tag{16}$$

An even ordered r-tuple of q-sets has total incidence qr, and every vertex in its union has incidence at least two. Its union therefore has at most $qr/2$ vertices. Since a multi-index α corresponds to $r!/\alpha!$ ordered tuples,

$$\begin{aligned}\sum_{\substack{|\alpha|=r\\ \alpha\text{ even}}}\frac{1}{\alpha!} &\leq \frac{1}{r!}\sum_{v\leq qr/2}\binom{N}{v}\binom{v}{q}^r\\ &\leq \left(\frac{qr}{2}+1\right)\left(\frac{qr}{2}\right)^{qr}N^{qr/2}.\end{aligned}$$

Using $\sigma_N^{2r} = ((q-1)!)^r N^{-(q-1)r}$ in (16) proves (14) after increasing D.

Let $m_N = \lceil A\log N\rceil$. Uniformly for $3 \leq r \leq m_N$, $D\log(r+1) = o(\log N)$. Hence, for all sufficiently large N,

$$\exp\big(Dr\log(r+1)\big)N^{-ar} \leq N^{-3ar/4}.$$

Summing from $r = 3$ to m_N gives $O(N^{-9a/4}) = o(N^{-2a})$. □

6 Completion of the modified FTW remainder estimate

Proposition 6.1 (Higher-chaos closure)**.** *For every fixed real β satisfying $0 < |\beta| < R_q$,*

$$\sum_{r\geq 3} \|P_r Y_N(\beta)\|_2^2 = o(N^{2-q}) = o(N^{-2a}). \tag{17}$$

Proof. Choose R with $|\beta| < R < R_q$ and put $t = |\beta|/R < 1$. Choose a constant A such that

$$2A|\log t| > a,$$

and set $m_N = \lceil A \log N \rceil$. Lemma 5.2 controls $3 \leq r \leq m_N$. For the remaining chaoses, (12) gives

$$\sum_{r>m_N} \|P_r Y_N(\beta)\|_2^2 \leq \frac{C_q}{1-t^2} N^{-a} t^{2(m_N+1)} = o(N^{-2a}).$$

Adding the two estimates proves the claim. □

7 External input II and the deterministic FTW coefficient

The remainder has now been controlled. To identify the deterministic coefficient $A_N(\beta)$ in (7), we use the published annealed free-energy limit. The derivative convergence needed below is not taken as an additional input; it follows from convexity.

Lemma 7.1 (Convex convergence and derivatives)**.** *Let $I \subset \mathbb{R}$ be open. Suppose that $h_N : I \to \mathbb{R}$ are finite, differentiable, convex functions and $h_N(x) \to h(x)$ pointwise, where $h \in C^1(I)$. Then $h_N \to h$ and $h_N' \to h'$ locally uniformly on I.*

Proof. Fix $[a,b] \Subset I$ and choose $u < a < b < v$ in I. Monotonicity of secant slopes bounds every secant slope of h_N on $[a,b]$ between quantities formed from $h_N(u), h_N(a), h_N(b), h_N(v)$. These endpoint values converge, so the h_N are equi-Lipschitz on $[a,b]$. Pointwise convergence on a finite mesh, followed by mesh refinement, gives uniform convergence there.

For $x \in I$ and sufficiently small $s > 0$, convexity gives

$$\frac{h_N(x) - h_N(x-s)}{s} \leq h_N'(x) \leq \frac{h_N(x+s) - h_N(x)}{s}.$$

First let $N \to \infty$ and then $s \downarrow 0$ to obtain $h_N'(x) \to h'(x)$. Finally, h_N' are nondecreasing. On a fine partition of $[a,b]$, monotonicity traps $h_N'(x)$ between its values at adjacent mesh points, while uniform continuity traps $h'(x)$ between the corresponding values of h'. Pointwise convergence at the finitely many mesh points then gives uniform convergence of the derivatives. □

Proposition 7.2 (High-temperature annealed free energy)**.** *There are $\beta_{\mathrm{FE}} = \beta_{\mathrm{FE}}(q) > 0$ and an even real-analytic function ℓ_q on $(-\beta_{\mathrm{FE}}, \beta_{\mathrm{FE}})$ such that, locally uniformly on this interval,*

$$\frac{1}{N} \log F_N(\beta) \longrightarrow \ell_q(\beta), \tag{18}$$

$$\frac{1}{N} \partial_\beta \log F_N(\beta) \longrightarrow \ell_q'(\beta). \tag{19}$$

Moreover,

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(\beta^3). \tag{20}$$

Proof. The pointwise convergence in (18) is Theorem 5 of [3], with the Gaussian–Rademacher universality result [4] used there, after the following normalization conversion. The Majoranas of [3] satisfy $\psi_i^2 = \mathbb{I}/2$, so $H_N^{\mathrm{GPSZ}} = 2^{-q/2} H_N$. If $u = 2^{q/2}\beta$, then symmetry of the Gaussian disorder removes the sign difference and

$$\frac{1}{N}\log F_N(\beta) = \frac{u}{N} f_{N,q}^{\mathrm{GPSZ}}(u), \tag{21}$$

where $f_{N,q}^{\mathrm{GPSZ}}(u) = u^{-1}\log \mathbb{E} Z_N^{\mathrm{GPSZ}}(u)$.

For completeness, we also extract the regularity and the first coefficient from the explicit formula in that theorem. In its notation,

$$\Phi_u(z) = \sum_{v\geq 1} \mu_v(u) z^v, \qquad u f_q^{\mathrm{GPSZ}}(u) = \Phi_u(t_*(u)) - 1 - \log t_*(u),$$

where $t_* \partial_z \Phi_u(t_*) = 1$. A contributing hypertree has $v = 1 + (q-1)s$ vertices and its coefficient contains

$$\rho(u)^{v-1}, \qquad \rho(u)^{q-1} = \frac{(q-1)!u^2}{2^{q+1}}.$$

The normal-convergence bounds in Proposition 3 of [3] therefore show that $\Phi_u(z)$ is analytic jointly in (u^2, z) near $(0,1)$. At $u = 0$, $\Phi_0(z) = z$ and $t_*(0) = 1$; the derivative with respect to t of $t\partial_z \Phi_u(t) - 1$ is 1 at $(u,t) = (0,1)$. The analytic implicit function theorem makes t_*, and hence the limiting right-hand side, analytic in u^2.

The first nontrivial term is the unique one-edge hypertree on q vertices:

$$\Phi_u(z) = z + \frac{u^2}{q 2^{q+1}} z^q + O(u^4).$$

It follows successively that

$$t_*(u) = 1 - \frac{u^2}{2^{q+1}} + O(u^4), \qquad u f_q^{\mathrm{GPSZ}}(u) = \frac{u^2}{q 2^{q+1}} + O(u^4).$$

Using $u = 2^{q/2}\beta$ in (21) gives (20).

It remains only to justify the two local-uniform assertions. Let $h_N = N^{-1}\log F_N$. If

$$\overline{\nu}_N(B) = \mathbb{E}\,\mathrm{tr}\,\mathbf{1}_{\{H_N \in B\}},$$

then $F_N(\beta) = \int e^{-\beta x}\overline{\nu}_N(\mathrm{d}x)$; hence h_N is finite, differentiable, and convex. Differentiation is justified by the Gaussian exponential-integrability bound in Lemma 1.2. The limiting function just obtained is C^1. Lemma 7.1 now proves both local-uniform convergences. □

By (7) and (19),

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2}\ell_q'(\beta). \tag{22}$$

8 The partition-function CLT

Theorem 8.1 (Partition-function CLT)**.** *Fix a real β satisfying*

$$0 < |\beta| < \min\{\beta_{\mathrm{FE}}, R_q\}, \qquad \beta\ell_q'(\beta) \neq 0.$$

Then

$$N^{q/2-1}\left(\frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)} - 1\right) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \xi \sim \mathcal{N}(0,1). \tag{23}$$

Moreover,

$$N^{q-2}\frac{\operatorname{Var} Z_N(\beta)}{(\mathbb{E}Z_N(\beta))^2} \longrightarrow \frac{q!}{2}\,\beta^2\ell_q'(\beta)^2, \tag{24}$$

and hence

$$\frac{Z_N(\beta)-\mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1). \tag{25}$$

Proof. Let

$$S_N = \frac{1}{\sqrt{2M_N}}\sum_{I\in\mathcal{I}_N}(g_I^2-1).$$

The classical Lindeberg theorem gives $S_N \Rightarrow \xi \sim \mathcal{N}(0,1)$. By Lemma 3.1,

$$Y_N(\beta)-1 = c_N(\beta)S_N + R_N(\beta), \qquad c_N(\beta) = \frac{\beta\,\partial_\beta \log F_N(\beta)}{\sqrt{2M_N}}, \tag{26}$$

where $R_N = \sum_{r\geq 3} P_r Y_N$ is orthogonal to S_N. Since $M_N \sim N^q/q!$, (19) gives

$$N^{q/2-1}c_N(\beta) \longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta).$$

Proposition 6.1 gives $N^{q/2-1}R_N \to 0$ in L^2. Slutsky's theorem proves (23). Orthogonality in (26) also gives

$$\operatorname{Var}(Y_N) = c_N(\beta)^2 + \|R_N(\beta)\|_2^2\,,$$

which proves (24). Finally, $\operatorname{Var}(Y_N) = \operatorname{Var}(Z_N)/(\mathbb{E}Z_N)^2$, and the assumed nonvanishing of the limiting coefficient allows one more application of Slutsky, proving (25). □

Remark 8.2. *Since $\ell_q'(\beta) = \beta/q + O(\beta^3)$ at the origin, the nondegeneracy condition in Theorem 8.1 holds for every sufficiently small real $\beta \neq 0$.*

Remark 8.3 (What replaces the non-uniform step in FTW)**.** *The complex second-moment estimate alone is only $O(N^{1-q/2})$ and is too weak for the CLT scale N^{2-q}. The gain comes from using it only for chaos orders above $A\log N$. Analyticity supplies the factor $(|\beta|/R)^{2A\log N}$ there, while the sign-gauge counting estimate is strong enough uniformly up to order $A\log N$. This is exactly the extra uniformity that the N-dependent test function $f_{N,\beta}$ requires beyond the fixed-test-function theorem of [1]. No CLT for $\log Z_N$ and no fluctuation-level assumption about the free energy is used.*

References

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